Fermionic quantum field theories as probabilistic cellular automata in three dimensions

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Wetterich has shown that certain probabilistic cellular automata in $1+1$ dimensions are exactly equivalent to fermionic quantum field theories, and posed the three-dimensional construction as an open problem. We construct one: a layered automaton (the "coincube") whose conversion blocks form the real quaternion representation of $\mathrm{Cl}(0,3)$, controlled by binary environment fields whose streaming outruns the carrier ("fresh tape")—no bit is ever consulted twice, and the quenched single-particle ensemble propagator equals the annealed Bloch operator identically at every time, a theorem of the model. Exact consequences: the step operator is unitary with respect to an explicit complex structure, with the spectral analysis its faithful complexification; the local Grassmann action is extracted mechanically; the single-carrier excitation is a Weyl fermion with an exactly isotropic leading cone, confirmed by gated measurement (slope ratios 1 within errors, the factorized-transport benchmark excluded by ≥ 10 error bars, helicity residues with chirality -1); at finite momentum the node carries a parity-odd anisotropy $c(q)\, k_x k_y k_z/|k|$, carried by every member of the four-component coin class (exhaustive enumeration) and cancelled exactly on the inversion-doubled Dirac branch, whose gap is exactly $2\arctan[q_m/(1-q_m)]$ while the off-resonant spectators of the computed 44-point Floquet–Weyl census stay exactly massless; a media-mediated interaction obeys $U_g = (1-2gq^2)\,U$ exactly at every cycle. The obstructions are charted alongside: ballistic transport admits only finite velocity sets (proven); Abelian sign gauges leave the factorized surface (measured over a finite gauge menu); no protected two-component node survives a table-exhaustive search. Finally, a dichotomy for the quaternionic event class: an isotropic cone forces overdamping—$Q = \omega_0/\Gamma \leq \pi/(2\ln 2)$ at every density, 0.66 at the working density (proven)—while in-state unitarity forces factorized transport, under stated hypotheses on the read geometry and the medium. The postselected two-boundary amplitudes evade the dichotomy, exactly unitary at all times under the fresh tape, with the positivity of their induced weight open: the exactly relativistic object of this construction is an amplitude whose probabilistic reading is not settled here.

I. INTRODUCTION

A probabilistic cellular automaton (PCA) is a classical statistical system: a lattice of bits, a probability distribution over bit configurations, and one local, invertible, homogeneous update rule applied in discrete time. Wetterich [1, 2] established that a class of such automata is *exactly* equivalent to fermionic quantum field theories in $1+1$ dimensions: the same wave function, density matrix, operators and expectation values, at every lattice size, with no approximation. The equivalence rests on three pillars: occupation bits map to fermions through a Grassmann functional integral; the step operator is a signed permutation, hence orthogonal; and a compatible complex structure turns orthogonal into unitary evolution. The construction in three or four spatial dimensions was left open—in Wetterich's own assessment, "[g]eneralizations of our formalism to three or four dimensions do not seem to encounter problems of principle. What is not yet achieved, however, are cellular automata that realize simple fermionic quantum field theories with Lorentz symmetry in three or four dimensions" [1]; the known four-dimensional route [3] transports Lorentz-invariant four-fermion composites rather than single fermions.

The program sits within a wider landscape. In the *unitary* quantum-cellular-automaton and quantum-walk tradition, discrete dynamics with postulated complex amplitudes yields Weyl and Dirac kinematics in one to three dimensions [6, 13–18], and Mlodinow and Brun have proven a no-go theorem for quantum-field limits of such automata above one dimension together with evasions that antisymmetrize distinguishable particles in two and three dimensions [19–21]; fermion doubling in these discrete settings remains an active subject [23]. The present work is in the *probabilistic* class, where amplitudes are not postulated but must emerge from classical statistics—the program of Refs. [1, 2, 8–11], with conceptual roots in 't Hooft's cellular-automaton interpretation [12]. The obstructions proven below (finite rays, Q-pinning) are properties of this probabilistic class; in the unitary setting, where complex amplitudes are postulated, the corresponding constraints do not arise in this form. The dressing and coin mechanisms below are the response to them.

This paper solves the construction half of that problem—the free single-fermion sector with exact relativistic kinematics at leading order—under the strictest contract: the propagating carrier is a single-fermion occupation number; the rule set (locality, invertibility with the unique-jump property, homogeneity, fermion-parity conservation per block, particle–hole symmetry, no hidden continuum input) is fixed at the outset and every construction below is verified against it mechanically. The Lorentz-symmetric half of Wetterich's sentence is *not* delivered beyond the leading cone: Sec. VI B proves that within this construction class the obstruction is structural—a parity-odd velocity anisotropy, of first order in the lattice spacing, tied to the chirality of the node—and quantifies the partial remedies. A single deliberate symmetry relaxation underlies this: a lone Weyl node is chiral, so the composite rule breaks spatial inversion, axis transpositions and time reversal by design, retaining the proper tetrahedral rotations and the antiunitary conjugation $U(-k) = U(k)^*$. The construction is exact in the following sense: the model's streaming schedule is chosen so that environment world-lines outrun the carrier's light cone, no bit is ever consulted twice, and the quenched single-particle ensemble propagator equals the Bloch operator theory identically at every time (Theorem 3); the closed forms of Secs. VI–IX are therefore statements about the automaton itself.

The results divide into obstructions and constructions, and we state the evidence class of every claim: [PROVEN] for symbolic or algebraic proofs, [MACHINE-CHECKED] for exact finite verifications (tolerances quoted), [MEASURED] for measurements, which are always accompanied by a known-answer gate run through the identical analysis pipeline. The complete verification code is public [29].

Obstructions (Sec. III): ballistic one-particle transport in this class has a *finite* set of group velocities, so no three-dimensional cone is possible (Theorem 1); on the symmetric vacuum all transport speeds are rational, while a tuned vacuum density q provides a continuous, exactly stationary knob with the closed dressing law $v = 2|1 - 2q|$; Abelian position-periodic sign gauges relocate band features but cannot bend the factorized ("diamond") group-velocity surface into a cone; and the only two-component isotropic nodes found are isolated fine-tuned points, every one frequency locked at $\omega_0 = \pi$ and without any dial (Theorem 2 and the accompanying exhaustive search).

Constructions (Secs. IV–IX): a four-component carried coin whose conversion blocks form the real quaternion representation of $\mathrm{Cl}(0, 3)$ produces an exactly isotropic Weyl cone; the automaton realizing it (the *coincube*) is a composition of manifestly legal layers whose fermionic lift is verified including all sign coherences; its fresh-tape streaming makes the quenched propagator exactly the operator theory (Theorem 3), with generic slower schedules and their re-read corrections analyzed alongside; the step operator is exactly unitary with respect to an explicit complex structure; the local Grassmann action is extracted in exact rational arithmetic; a dense-sweep census with computed topological charges charts the full gapless spectrum; the node's finite-momentum anisotropy is derived, proven unavoidable at four components by exhaustive enumeration, and cancelled exactly on the doubled branch (Sec. VI B); spatial-inversion doubling yields a Dirac fermion with an exact, continuously tunable mass for the resonant node quartet; and a media-mediated interaction with a continuous coupling is certified, its damping law exact at every cycle.

Finally, Sec. X proves the second principal result: a dichotomy. Within the quaternionic event class, an isotropic cone *forces* a nonzero, exactly mode-blind visibility decay (the annealed step operator factorizes as a scalar times a unitary, Proposition 1), bounded by $\kappa = \pi/(2 \ln 2)$ on the node's phase advance per unit decay and leaving the in-state excitation overdamped ($Q = \omega_0/\Gamma \approx 0.66$ at the working density); while in-state unitarity forces factorized transport and destroys the cone. Correlated media are included; outside the event class the bound provably fails at an exhibited boundary. The two-boundary (in–out) propagators—postselected amplitudes, legitimate objects in the same transfer-matrix formalism—evade the dichotomy, under the fresh tape at all times: they form a one-parameter unitary family with the cone intact, and the square-root boundary is proven to be its unique medium-blind member.

Throughout, the lattice spacing and the duration of one full update cycle are unity; frequencies are per cycle and momenta are in lattice units.

II. SETUP

A. States, step operator, lifts

Configurations τ are assignments of occupation bits to species at the sites of a cubic lattice. The probabilistic information is a real wave function q_τ with $p_\tau = q_\tau^2$ [1]. One time step applies a deterministic, invertible map of configurations together with a sign gauge: the step operator $\hat{S}$ is a *signed permutation* of the configuration basis, hence real orthogonal. A rule is *legal* if it is a composition of layers, each of which is (i) local, (ii) a bijection on configurations with a unique image (the unique-jump property), (iii) homogeneous, and (iv) fermion-parity even on its support, so that a fermionic lift exists. The lift assigns the signs: occupation bits map to fermionic modes in a fixed Jordan–Wigner ordering, and each layer's signed permutation must equal the matrix of an explicit fermionic operator in the occupation basis. All lifts below are verified at this level, including two-particle sign coherence (Sec. IV).

B. Environment fields and dressed carriers

Besides the carrier species, the automata below contain autonomous binary *environment* fields: one per spatial axis (density q), plus one for the mass sector (q_m) and one for the interaction (g). Environment dynamics is free streaming by pair-swap layers, applied in fast phase-continuing batches—the *fresh-tape* schedule defined in Sec. IV and used in Sec. V; a Bernoulli product measure is exactly stationary under any pair-swap schedule. Carriers never modify the environment except through the explicitly constructed interaction layer of Sec. IX.

The vacuum is the product of the carrier vacuum with the Bernoulli environment measure: translation invariant, clustering, with a continuous parameter q that is the central dial of the construction.

III. OBSTRUCTIONS

Theorem 1 (Finite rays) *For any legal rule with finitely many species per site that conserves particle number in the one-particle sector—the sector with exactly one occupied bit across all species on an otherwise empty lattice—the one-particle Bloch operator is a monomial matrix for every momentum k. Consequently every dispersion branch is exactly linear in k, the set of group velocities is finite, and no three-dimensional cone $\omega = v|k|$ is possible ballistically.* [PROVEN]

The proof is elementary—a signed permutation with momentum phases has monomial matrix elements, and eigenvalues of monomial matrices are roots of single monomials—but the consequence is structural: the error is scale invariant, so no continuum limit removes it, and periodic cycles of rotated rules do not evade it because products of unique-jump operators are unique-jump. In one spatial dimension the light cone genuinely consists of two rays, which is why the 1+1-dimensional construction exists. An adversarial test suite (seven evasion strategies with re-seeded variants, plus reconstruction tests that fit the Bloch operator from real-space propagation rather than constructing it) accompanies the theorem in the repository. The same suite verifies how the construction of this paper relates to the theorem: at fixed media the coincube cycle is a signed permutation of the (site, channel) basis but not homogeneous, so the theorem does not apply per realization—and indeed every realization retains a finite velocity set. The physical configuration carries $O(qL^3)$ occupied environment bits, so the propagating object is a carrier dressed by an extensive environment, not the theorem's one-particle state, and the cone exists only in the media-averaged propagator, which is not monomial. The construction evades the theorem's hypotheses, not its conclusion.

Two further results constrain the construction. First, on the symmetric (density-$\frac{1}{2}$) vacuum the transport speeds of the autonomous-streaming subclass of conditional rules [2] are rational [PROVEN] (frozen-vacuum and fully coupled rules are outside this statement); the vacuum density q is the legal continuous parameter, with the closed dressing law, for the representative rule used throughout,

$$v(q) = 2\,|1 - 2q|, \tag{1}$$

proven as a two-class Markov-additive limit and verified against exact ring enumeration as polynomial identities in q [PROVEN]. Second, dressing alone yields a factorized walk whose group-velocity surface is the ℓ^1 ball (the *diamond*): slope ratios along (110) and (111) relative to (100) equal $\sqrt{2}$ and $\sqrt{3}$. Abelian position-periodic sign gauges (Kogut–Susskind axis phases [4, 22], per-stall signs, and their products) relocate the nodal points and modify residues but leave the ratios on the diamond, measured as $r_{110}/r_{111} = 1.40/1.68$ (base and stall gauges, identical spectra) and 1.46/1.82 (KS gauges) [MEASURED]—an early ungated instrument whose absolute anchor deviates 3.5–5.5% from the exact operator, quoted only for the on-diamond pattern; a carried non-Abelian structure is therefore necessary.

Theorem 2 (Two-component exclusion) *Consider legal automata whose carrier has a two-dimensional internal space: per-axis cycles of the monomial factors $(1-p)E_a + pC_a$ or $E_a[(1-p)\mathbb{1} + pC_a]$, where $E_a = \mathrm{diag}(e^{\pm ik_a})$ and C_a is a real signed permutation, with one or two (blocked) engagements per axis. At every conversion weight $p \neq \frac{1}{2}$ the single-engagement cycles have no degenerate propagating pair at any momentum (rotation coins force $(1-p)^2 = p^2$; reflection coins never produce one; diagonal coins degenerate).* [PROVEN]

The remaining cases are closed by search—exhaustive in the discrete tables, heuristic in momentum, sampled in weight—not by proof. The only isotropic two-component Weyl nodes found in this class are *isolated fine-tuned points*: the single-origin placement at exactly $p = \frac{1}{2}$ ($\omega_0 = \pi$, $v = 1$, $\chi = -1$, $|\lambda_0| = 2^{-3/2}$, all forced, width $\Gamma = \frac{3}{2}\ln 2$ at the chord midpoint—the maximally damped point of Theorem 4) [PROVEN]; and the blocked two-origin cycles at numerically located tuned weights ($p^* = 0.3300$ additive, $p^* = 0.3204$ multiplicative; anisotropy zero within extrapolated residuals $\leq 8.5 \times 10^{-4}$), both likewise locked at $\omega_0 = \pi$ and heavily damped, with computed chiralities $\chi = -1$ and $\chi = +1$ respectively [MACHINE-CHECKED]. At the sampled weights $p \in \{0.15, 0.25, 0.35, 0.5\}$ every design has a strictly positive anisotropy floor [MACHINE-CHECKED]; the floors necessarily dip toward zero on approach to the tuned p^*, which lie between the sampled weights. (In the *unitary* walk setting, where amplitudes are postulated, two-component three-dimensional constructions exist [27, 28]; the exclusion here is specific to the probabilistic class.)

Two-component constructions therefore admit no *protected* relativistic carrier: every isotropic node found is an isolated point in parameter space, its quasienergy locked at $\omega_0 = \pi$—no mass dial, no width dial, no vacuum-density dressing. The minimal internal space with a symmetry-protected node on an entire parameter family is four-dimensional and real, where the Clifford algebra $\mathrm{Cl}(0,3)$ has its quaternionic representation; that is the construction of the next section.

IV. THE COINCUBE AUTOMATON

A. Layers

The carrier has four species per site: a two-bit coin (b_1, b_2), channel index $c = 2b_1 + b_2$. Writing X, Z for the real Pauli matrices and XZ for their (antisymmetric) product, define the conversion triple and direction assignments

$$\begin{aligned} C_x &= XZ \otimes \mathbb{1}, & d_x &= \mathrm{diag}(Z \otimes \mathbb{1}), \\ C_y &= Z \otimes XZ, & d_y &= \mathrm{diag}(Z \otimes Z), \\ C_z &= -X \otimes XZ, & d_z &= \mathrm{diag}(\mathbb{1} \otimes Z). \end{aligned} \tag{2}$$

The C_a are pairwise anticommuting with $C_a^2 = -\mathbb{1}$: the real quaternion representation of $\mathrm{Cl}(0,3)$ [MACHINE-CHECKED]. Each full cycle applies, for each axis a and each of two block origins, three layers:

L2 (conversion). At every site, controlled on the axis-a environment bit at that site, the two channel pairs of C_a are swapped (a one-site block; two bit flips, parity even).

L1 (motion). Channel c translates by $d_a(c) \in \{\pm 1\}$ along axis a, unconditionally.

L3 (environment). The axis-a environment field streams along $s_a = a{+}1 \pmod 3$: each axis-a substep applies *three* consecutive pair-swap layers to the field, with origins that continue the field's global alternation phase (the n-th swap ever applied has origin $n \bmod 2$, counted across substeps and cycles). Each layer is the same certified environment layer; the schedule is deterministic, homogeneous and autonomous. Phase continuation is required: a schedule restarting each batch at origin 0 repeats an origin at the batch boundary and the repeated pair cancels (measured transport 0 per batch, versus exactly 3 for the phase-continuing batch) [MACHINE-CHECKED]. Under this schedule every environment bit moves at exactly ± 6 sites per cycle along s_a, in a direction fixed at $t = 0$ by its parity class, while the carrier moves at most 2 sites per cycle per axis—the environment outruns the carrier, with the consequences of Sec. V. Two design constraints are mandatory. *Cross-streaming* ($s_a \neq a$): if the field streams along its own axis, a co-moving carrier re-reads its own bit, conversions arrive in nearly scalar bursts ($C_a^2 = -\mathbb{1}$), and the measured Weyl pair splits into one damped-real and one oscillating pole—an exceptional-point transition of the damped band structure in the sense of non-Hermitian topology [24]—and the node is destroyed [MEASURED]. The second is even linear size L (enforced in code); all statements below assume even L.

Each layer is manifestly legal; the composition defines the automaton.

B. Fermionic lift and sign coherence

The lift of L2 is an environment-controlled pair of $\pm 90^\circ$ Givens rotations $G = \exp[\frac{\pi}{2}(a_c^\dagger a_{c'} - a_{c'}^\dagger a_c)]$ acting on same-site modes: a signed permutation of the Fock basis, parity-even, identity at empty control, with single-particle matrix equal to the corresponding signed-swap block of C_a and $G^2 = -\mathbb{1}$ on the one-particle sector; the quaternion sign structure *is* the fermionic rotation sign [MACHINE-CHECKED] (dense verification, all axes). Translations and environment swaps lift with the standard reordering parity. The composite cycle is sign coherent: in the segregated mode ordering (all carrier modes before all environment modes) the full lifted cycle is a signed permutation whose one-particle sector equals the stated rule and whose two-particle sector equals the exact antisymmetrized square $\Lambda^2 M_1$—verified in genuine three-dimensional geometry on all 5778 pair states at linear size $L = 3$, with mutation controls (deliberately corrupted sign rules fail the check) [MACHINE-CHECKED]. At $L = 2$ the two sub-steps per axis make net translation crossings even, so $L = 2$ is blind to translation-sign errors; three-dimensional sign claims must be checked at $L \geq 3$.

V. THE FRESH-TAPE PROPERTY

The environment schedule of L3 was chosen so that no bit is ever consulted twice:

Theorem 3 (Fresh tape) *Consider the coincube in the single-particle sector with the L3 schedule of Sec. IV and the Bernoulli(q) product vacuum. (a) On the infinite lattice, no conversion history reads the same physical environment bit twice, at any time. Consequently the quenched ensemble propagator equals the annealed propagator* identically, $G_{\mathrm{quenched}}(t) = G_{\mathrm{annealed}}(t)$ *for all t: every closed form derived from the annealed Bloch operator $U(k)$—the node census, the propagator factorization, the linear conversion law, the mass law—is exact for the automaton itself. (b) On the L^3 torus (L even) the same holds for every evolution of $T \leq \lceil L/8 \rceil$ cycles, and this horizon is sharp: the first wrap-induced re-read joins two reads separated by exactly $2\lceil L/8 \rceil$ of the field's own axis-a substeps.* [PROVEN], [MACHINE-CHECKED]

The proof (Appendix A) is a kinematic argument on three finite lemmas; every lemma and the sharpness of (b) are machine-checked, and the equality in (a) is verified as an exact path sum over *all* conversion histories, with physical bit labels tracked through the real streaming permutations: all four launch channels at $T = 3$ (2^{18} histories each, deviation 2.8×10^{-17}) and channel 0 at $T = 4$ (all 2^{24} histories, deviation 9.1×10^{-17}) [MACHINE-CHECKED]. The scope is every observable linear in the evolved field—amplitudes, propagators, and the spectra derived from them; quadratic estimator statistics (variances of paired-media estimators) still distinguish quenched from annealed ensembles, and nothing in the theorem concerns them.

A. Generic schedules and re-read corrections

The fresh-tape property belongs to the schedule, not to the layer type. The generic single-swap schedule (one pair-swap layer per axis substep, origins alternating per substep) transports bits at ± 2 sites per cycle on the two parity sublattices—pair-swap streaming counter-propagates; it is not a unit shift—which exactly matches the carrier's maximal transverse speed: carrier and bit can co-move, and the same bit is consulted again after one or two cycles. In the exact $T = 3$ path sum this produces 13 654 re-read branch events and a quenched-vs-annealed propagator deviation of 1.7×10^{-3}; a separate twenty-cycle kinematic certificate counts 380 re-read pair classes for this schedule against zero for the fresh-tape schedule [MACHINE-CHECKED]. At production scale the re-read corrections are directly measurable: under the generic schedule the quenched node parameters shift from the operator values by +2.8% ($|\lambda_0|$) and +8.6% (ω_0) at $q = 0.08$ (of which only +0.3%/+2.3% is instrumental, from the gate row), and the interaction damping law acquires excursions beyond the first cycle (2–2.5% at $g = 0.3$, up to 6% at $g = 0.6$; measured at $L = 12$ inside the generic run's own wrap regime, so re-read and torus-wrap contributions are not separated there) [MEASURED]. Under the fresh-tape schedule every one of these corrections vanishes identically (Theorem 3); the measured comparison is Fig. 1.

VI. FREE SPECTRUM: THE WEYL CONE

A. Closed forms

By Theorem 3 every engagement reads a fresh Bernoulli(q) bit, so the one-particle transfer per substep is exactly $T_a(k) = E_a(k_a)\,[(1-q)\mathbb{1} + qC_a]$ with $E_a = \mathrm{diag}(e^{ik_a d_a})$, and the cycle operator is $U(k) = T_z(k_z)^2 T_y(k_y)^2 T_x(k_x)^2$ (axes applied in the order x, y, z; the closed forms below hold for either orientation of the cycle, the census numbers are specific to this one): the operator theory of this section is the automaton's single-particle dynamics, not an idealization of it. (The same operator arises for any schedule under independent per-engagement reads—the annealed average—which is how the generic schedules of Sec. V A are compared against it.)

Proposition 1 (Scalar–unitary factorization) *Each conversion block C_a is real, antisymmetric and orthogonal, so $T_a(k)^\dagger T_a(k) = [(1-q)^2 + q^2]\,\mathbb{1}$ identically in k; hence*

$$U(k) = \rho^6\, V(k), \qquad \rho^2 = (1-q)^2 + q^2, \tag{3}$$

with $V(k)$ exactly unitary for every k. [PROVEN]

Proof: $(\alpha\mathbb{1} + \beta C)^\dagger(\alpha\mathbb{1} + \beta C) = (\alpha^2+\beta^2)\mathbb{1} + \alpha\beta(C + C^T) = (\alpha^2+\beta^2)\mathbb{1}$; as a machine check, every eigenvalue modulus equals ρ^6 to 10^{-15} over random momenta at four values of q [MACHINE-CHECKED]. Two closed forms are immediate: the node modulus $|\lambda_0|^2 = \rho^{12} = [(1-q)^2 + q^2]^6$, and the exact periodicity $U(k + \pi e_a) = U(k)$, since $E_a(k_a + \pi) = -E_a(k_a)$ and the sign cancels in T_a^2 [PROVEN]. The linear splitting of the node doublet is proven symbolically to be exactly isotropic, identically in q, with speed $v(q)$ rising from $2/\sqrt{3}$ at $q \to 0$ to a maximum 1.207 near $q = 0.30$ ($v(0.08) = 1.1806$) [PROVEN]; numerically the splitting isotropy holds to 2×10^{-4} over 50 random directions after $h \to 0$ extrapolation [MACHINE-CHECKED].

B. The chiral anisotropy

The exact isotropy of the node is a property of the leading order alone. The splitting is

$$s(k) \;=\; 2v\,|k| \;+\; c(q)\,\frac{k_x k_y k_z}{|k|} \;+\; O(|k|^3), \tag{4}$$

the second term proportional to the lowest anisotropic invariant of the chiral tetrahedral group: to this order the slope depends on the direction only through $\hat{k}_x\hat{k}_y\hat{k}_z$ (exactly, for A_4-related directions; to $O(|k|^4)$ otherwise), and $c = 42.7, 20.0, 10.5, 8.6$ at $q = 0.08, 0.15, 0.25, 0.30$, growing toward small density [MACHINE-CHECKED]. The quaternionic triple carries an orientation ($C_x C_y C_z = +\mathbb{1}$, so the composed cycle order gives $C_z C_y C_x = -\mathbb{1}$), so no signed coin relabeling implements an odd axis permutation; the spectral symmetry group is exactly $T \cong A_4$ (Sec. VI C), and A_4 admits the degree-3 invariant $k_x k_y k_z$: the algebra that forces the isotropic leading cone forbids every symmetry that would exclude the odd term. The term is a property of the entire class: the 4×4 signed-permutation group contains exactly 12 elements squaring to $-\mathbb{1}$ (the left- and right-multiplication quaternion families) and 48 ordered anticommuting triples, and over every triple, signed axis assignment, block ordering and balanced direction-table combination that preserves an isotropic-leading node, the odd coefficient never falls below 0.85 of the production value [MACHINE-CHECKED]. Legal deterministic counterterm layers—coin-conditional transverse translation dipoles, which read no environment bits and leave Theorem 3 untouched—couple to the correct tensor structure with bounded strength: the best family member removes 45% of the coefficient, with the leading isotropy intact and the residual still of rank 3 [MACHINE-CHECKED].

Three consequences are quantitative. *Scale*: the relative velocity anisotropy over the sphere is $c\,|k|/(3\sqrt{3}\,v) \simeq 7.0\,|k|$ at $q = 0.08$, of order one at the smallest momentum of the $L = 48$ production torus, and below 1% only for $L \gtrsim 4400$: the isotropic cone is a $k \to 0$ statement, and the approach to it is one power of the lattice spacing slower than for a factorized lattice fermion. *The instrument*: the cone estimator of Sec. VI D pools every

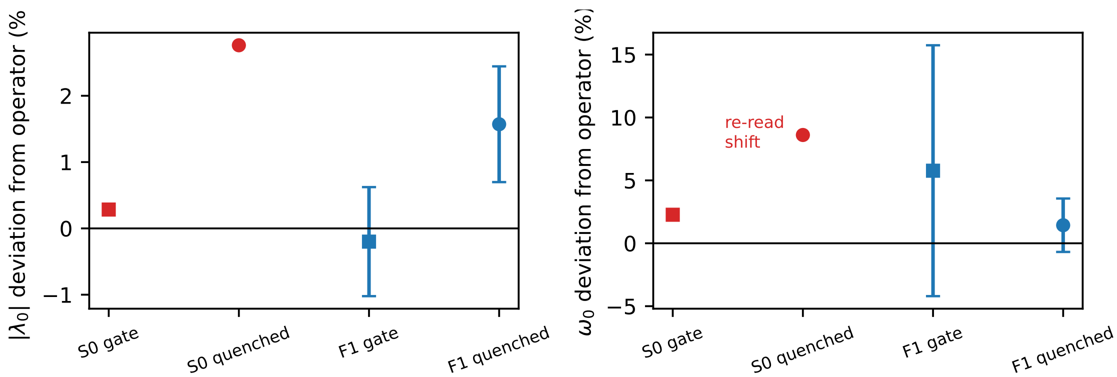

FIG. 1. The fresh-tape identity at scale: measured quenched node modulus (left) and frequency (right) as deviations from the exact operator values, under the generic schedule (red; re-read shifts, gate rows shown for the instrumental part) and under the fresh-tape schedule (blue, with block-jackknife errors). The generic-schedule points predate per-node jackknife storage and are shown as central values; their instrument's offsets are the gate points. [MEASURED]

direction with its full $\pm$ star, and the star average annihilates each odd term identically [PROVEN] (verified to 2×10^{-17} [MACHINE-CHECKED]), so the measured ratios are statements about the even sector and the $k\to0$ slope; the direction-resolved ratios of the exact operator at the probe radii are $r_{111} = 1.116, 1.207, 1.366$ at $\delta = 0.03, 0.05, 0.08$. *Cancellation*: the term is odd under spatial inversion, and the inversion-doubled model of Sec. VIII cancels it exactly, at every momentum and mass density. The term is tied to the net chirality: an eight-component chirality-2 construction with full proper-octahedral covariance (two same-orientation blocks exchanged by a legal block swap) removes it from the dispersion only by moving it into the splitting of a nested double cone that no coupling can close [MACHINE-CHECKED]; it vanishes at first order precisely when the net chirality does.

C. The gapless census

The gapless set of the unitary factor $V(k) = U(k)/\rho^6$ is charted by a dense sweep of the reduced zone $[0,\pi)^3$: minimum pairwise eigenvalue distances on a 64^3 grid, Nelder–Mead refinement of every local minimum below 0.15 to machine precision, sphere-probe classification of each refined degeneracy (an isolated point keeps a finite gap in all directions; a line member has an antipodal zero pair), and orbit grouping under the spectral symmetry group. That group is determined empirically: of the 48 signed coordinate permutations, exactly the 12 proper rotations of the chiral tetrahedral group $T\cong A_4$ preserve the spectrum, and the 12 complementary operations (including $k\to-k$) map it to its conjugate—no odd permutation appears, reflecting the cyclic $x\to y\to z$ layer ordering. For every isolated node the chirality is *computed* as a Chern number: the Fukui–Hatsugai lattice Berry flux of the upper doublet member over an enclosing sphere, built from the spectral projectors of V (well defined by Proposition 1) and calibrated on $\exp(+ik\cdot\sigma)$; every quoted charge is an exact integer, stable under two sphere radii and two angular grids [MACHINE-CHECKED].

TABLE I. The gapless census of the coincube walk (chord weights, $q = 0.08$): five point orbits (44 isolated Weyl points), three nodal lines, and their triple junction. Quasienergies are +Im-branch values; charges are computed Chern numbers per node.

orbit	rep. k/π	ω/π	mult.	χ_+/χ_-
corner	$(0,0,0)$	0.1006	1	-1 / $+1$
half-points	$(\frac{1}{2},0,0)$+perms	0.9221	3	$+1$ / -1
$\langle111\rangle$ octet A	$(0.265, 0.265, 0.735)$	0.5142	8	-1 / $+1$
$\langle111\rangle$ octet B	$(0.243, 0.243, 0.757)$	0.5652	8	$+1$ / -1
generic orbit	$(0.028, 0.401, 0.902)$	0.9479	24	-1 / $+1$
nodal lines	$(t,\frac{1}{2},\frac{1}{2})$+perms	t-dep.	3	—
R junction	$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$	1	1	$U=-\rho^6\mathbb{1}$

Table I is the census. Beyond the corner and the half-points the zone carries 40 further Weyl points: two $\langle111\rangle$ octets with members $(\pm t,\pm t,\pm t)$ mod π and a fully generic 24-point orbit. Multiplicities count distinct momenta; the corner and half-points host both branches at each momentum, while the octets and the generic orbit host one band-touching event per point, so the per-branch event counts are $(1,3,4,4,12)$ and the point charges on the +Im branch sum to $S_+ = -1+3-4+4-12 = -10$ (and $S_- = +10$)—a bookkeeping total over the +Im quasienergy window, which is a convention, not a band; the invariant statement is the conjugation pairing below. Every other local minimum of the grid-basin search refines to a nonzero gap (smallest avoided crossing 0.019 at $q = 0.08$, 0.050 at $q = 0.15$, and 0.0013 at $q = 0.30$—the weakest completeness margin, at the density where the topology itself changes): to those resolutions the lists are complete. The same five-orbit topology with identical charges holds at chord $q = 0.15$ and at the two-boundary (arc) weights at $q = 0.08$—the extra species are not artifacts of the damping factorization—but it is not permanent in q: the generic orbit annihilates pairwise on the $\{k_a = 0\}$ planes at $q_c \in (0.292, 0.294)$, leaving a 20-point census with $S_+ = +2$ at $q = 0.30$ [MACHINE-CHECKED].

Two facts govern the charge balance. First, the eigenphases of $V(k)$ cover the entire circle—a 360-bin occupancy histogram over a 40^3 momentum grid has no empty bin—so V has no Floquet quasienergy gap anywhere: the static Nielsen–Ninomiya [5] argument applies to no branch or quasienergy window, while the Floquet-extended zero-sum rule [25, 26], which fixes the total chirality of a gapless unitary at its bulk winding number $W_3[V]$, is satisfied identically: all layers are real, so $V(-k) = V(k)^*$, under which W_3 is odd, hence $W_3[V] = 0$ [PROVEN]—precisely the conjugation-pairing total below. $S_+ = -10$ violates nothing. The one exact global constraint is that pairing: $U(-k) = U(k)^*$ pairs every event (k, ω, χ) with $(-k, -\omega, -\chi)$—verified event by event—and the two-branch total vanishes identically. Second, the nodal lines: the loop Berry phase around a line is close to π but *not* quantized (0.954π–0.998π along the line; no protecting symmetry was identified), and no closed Gauss surface enclosing one line while avoiding the gapless set exists, since the three lines intersect at the R point; the loop-phase winding along the full line is consistent with zero at the smaller tube radius but is radius-unstable, so no monopole charge is assigned to the lines—and none is needed: the only surviving global constraint, $W_3[V] = 0$, is already saturated by the point charges' conjugation pairing [MACHINE-CHECKED].

All corner-local measurements below probe the doublet at $k = 0$, $\omega_0 = 0.1006\pi$. The nearest other gapless feature of any kind lies at torus distance 0.4134π from the corner (a generic-orbit member; the octets follow at 0.4216π and 0.4590π, the half-points at $\pi/2$, the nodal lines at 0.7071π), and the nearest spectator point node in quasienergy is 0.414π away, so corner-local fits at $|k| \lesssim 0.15$ are separated from every spectator by more than two decades in momentum. But the walk is a lattice fermion with a larger spectator census than a corner count suggests: its additional species are displaced, not absent [23].

D. Quenched measurement

On the streaming (quenched) environment the propagator is measured from the exact per-medium single-particle signed field, averaged over media. The instrument fits the full 4×4 one-cycle Bloch matrix per momentum from all four basis launches, pools symmetry-equivalent momenta at the level of characteristic-polynomial coefficients (basis invariant, hence unbiased) before rooting, probes momenta off the lattice grid inside the node's linear range, and extrapolates $\delta k \to 0$. The $\pm$-star pooling annihilates the odd anisotropy of Sec. VI B identically, so all ratios below probe the even sector and the $k \to 0$ slope. Every production run carries an annealed known-answer row through the identical pipeline; a run whose gate row fails is discarded. Near-degenerate pole extraction is the dominant threat (eigenvalues of a noisy nearly defective matrix split as $\sqrt{\text{noise}}$), which this design removes.

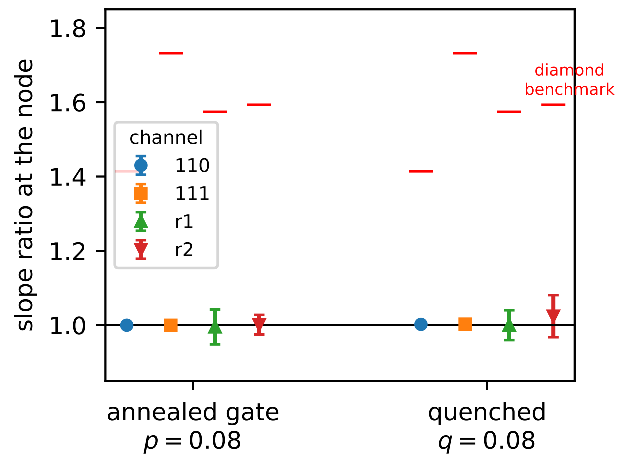

FIG. 2. Slope ratios at the node, annealed gate row and quenched row, against the factorized-transport benchmark. Error bars are leave-one-block-out jackknives over ten independent medium blocks; by Theorem 3 the two rows measure the same operator. [MEASURED]

Results (Fig. 2): the annealed gate row reproduces the exact operator (cubic-channel ratios within 10^{-3} of unity; off-symmetry channels within their own statistics; node modulus 0.619 ± 0.005 vs 0.620, frequency 0.334 ± 0.032 vs 0.316), and the quenched row lands on the same operator, as Theorem 3 demands: node modulus 0.6300 ± 0.0054 ($+1.6\%$, 1.8σ; an independent-seed replication at doubled R returns 0.6180 ± 0.0038, -0.4%—the excursion is statistical) and frequency 0.3205 ± 0.0067 ($+1.4\%$, 0.7σ) against the closed forms, with slope ratios $r_{110} = 1.002\pm0.002$, $r_{111} = 1.003\pm0.003$, $r_{\mathrm{a}} = 1.000\pm0.040$ and $r_{\mathrm{b}} = 1.024 \pm 0.057$ for the two off-symmetry directions. The factorized-transport (diamond) benchmark is excluded in every channel—an exclusion of the benchmark *value*, not of a competing automaton, since a fully factorized walk supports no measurable doublet at all (Sec. X): the weakest exclusion—by construction the off-symmetry channel with the fewest orbit members—sits 10.0 of its error bars from the diamond value, and the cubic channels sit more than 160 error-bar widths away. The estimator's power to resolve anisotropy is established by the positive controls: end to end on the off-symmetry channels (a legal broken-coin walk under the same schedule, recovered within errors, anisotropy resolved at $8.6\sigma/11.6\sigma$), and for the cubic stars—which orbit pooling symmetrizes for cubic-breaking truths—against a synthetic diamond-splitting truth through the identical analysis. Absolute slopes carry the large medium-ensemble and fit-window uncertainty seen throughout (± 0.18 at this scale); the two rows scatter around the operator value 1.18 accordingly, and the node parameters and ratios are the precision observables. [MEASURED]

E. Helicity residues

At the node the effective generators h_a on the doublet satisfy the exact Clifford algebra $\{h_a, h_b\} = 2v^2\delta_{ab}$; because the node frame of the damped (non-normal) operator is not unitary, the Pauli coefficients of h_a are complex with *bilinear* orthogonality $R^T R = v^2 \mathbb{1}$ ($R \in O(3, \mathbb{C})$), and $\det R = -v^3$ exactly: chirality $\chi = -1$ [MACHINE-CHECKED]. The split branches' spectral projectors equal the helicity projectors $\frac{1}{2}(\mathbb{1} \pm n \cdot \sigma)$, $n = Ru/\sqrt{(Ru)\cdot(Ru)}$. Measured on the quenched vacuum (the 26 directions of the cubic $\{100\}/\{110\}/\{111\}$ stars at each of the three—by π-periodicity identical—corner representatives; gate row passed): chirality -1, stable under every jackknife resampling in both rows; pointwise helicity-direction agreement $|1 - n \cdot n_{\text{exact}}|$ of mean 0.010 ± 0.002, with the gate row returning 0.013 ± 0.008 on the same operator. The bilinear isotropy statistic (a maximum over the nine entries of $R^T R/s - \mathbb{1}$, hence positively biased) is reported only as an upper bound: the quenched row gives 0.060 ± 0.018 against the gate row's 0.065 ± 0.017—the two rows are statistically identical, as Theorem 3 demands, and both sit at the instrument's noise floor, bounding the node's bilinear anisotropy at $\lesssim 0.1$ [MEASURED]. (The residue projectors are rank one by construction of the spectral estimator, so "purity" is not reported as a measurement.) One convention is essential for eigenvector work: the $e^{-ik\cdot x}$ Fourier contraction estimates $U(-k) = \bar{U}(k)$, whose upper-half eigenvectors belong to the opposite-helicity conjugate doublet; residue measurements must use $e^{+ik\cdot x}$.

VII. EXACT QUANTUM MECHANICS

A. Complex structure

Let P be the lifted carrier particle–hole conjugation, the Majorana string $\prod_i (a_i + a_i^\dagger)$ over carrier modes. In the segregated ordering, P is real orthogonal with $P^2 = +1$ (carrier mode number $M = 4L^3 \equiv 0 \bmod 4$), and

$$[\hat{S}, P] = 0 \tag{5}$$

exactly, including all lift signs, as an operator identity on the full Fock space at every lattice size [PROVEN]: a string-factorization lemma reduces each layer's commutation with the Majorana string to a fixed finite block check (conversion and imprint blocks act on same-site modes, so their Jordan–Wigner spectators are block internal), and the translation lift contributes $\operatorname{sign}(\pi) = (-1)^{(L-1)L^2} = +1$ for every L (Appendix A). Independent sector certificates confirm the identity directly through the three-carrier sectors at $L = 2, 3$ (204 156 three-carrier states at $L = 3$, 1.6×10^6 state–draw evaluations; at odd L, where the even-L pair-swap batch is undefined, the certificate streams by a single plane transposition—the sign identities do not depend on the schedule), the $M-2$ sector by particle–hole duality, and one- and two-particle sectors at $L = 4$, over eight environment draws [MACHINE-CHECKED]. With the grading $\eta = \operatorname{sign}(N_c - M/2)$—conserved by every layer, odd under P, and size independent away from half filling—define

$$K = \operatorname{diag}(\eta), \qquad I = P \operatorname{diag}(\eta). \tag{6}$$

Then $K^2 = 1$, $I^2 = -1$, $\{K, I\} = 0$, I antisymmetric, and $[\hat{S}, I] = 0$; with multiplication $(\alpha + i\beta)q := \alpha q + \beta I q$ and the induced Hermitian form, $\hat{S}$ is a unitary operator on a complex Hilbert space [1]: the automaton's quantum mechanics is exact [MACHINE-CHECKED]. On the substrate the construction extends to the half-filled sector by an explicit orbit-wise grading; in three dimensions no size-independent grading of the half-filled sector was found and the question is left open. Away from half filling, all carrier sectors are covered at every size, by proof and by direct verification.

B. Bridge to the spectral analysis

The complex structure above is global (a Majorana string), and the spectral results of Sec. VI use the ordinary Fourier i. These are the same quantum mechanics, by an explicit intertwiner: the map $\Phi(u+iv) = u \oplus (-Pv)$ is a complex-linear isomorphism from the complexified one-particle sector with the Fourier i onto the $(V_1 \oplus V_{M-1}, I)$ eigensector of the automaton's Hilbert space, intertwining the dynamics (this is exactly $[\hat{S}, P] = 0$ restricted to the sector) and the Hermitian forms. Consequently the measured propagator *is* an I-Hermitian matrix element, every $\pm\omega$ conjugate pair of the Fourier analysis appears once as a particle and once as an antiparticle in the (K, I) picture, and the chirality bookkeeping of Sec. VI is the particle/antiparticle structure of a single Weyl node. Two scope remarks are owed. The complex structure is nonlocal—a Majorana string mapping k-particle to $(M-k)$-particle sectors—so no local occupation observable commutes with I; the identification of measured objects with I-Hermitian matrix elements is established here for the one-particle sector, and multi-carrier correlators (such as the C_2 certificates of Sec. IX) are not re-expressed in the I picture. The bridge's content follows from $[\hat{S}, P] = 0$, which is proven at every size (Appendix A); the direct machine check—at $L = 3$, a sign-sensitive size (Sec. IV), over three environment draws including the working density—confirms the intertwining identities exactly (deviation zero at machine level) and the form isometry to 9×10^{-15} [PROVEN], [MACHINE-CHECKED]. What remains genuinely distinct is the damping: the in-state reduced dynamics is a contraction (the chord), and only the two-boundary amplitudes of Sec. X are unitary (the arc); the bridge identifies the kinematics, not the dissipative structure.

C. Grassmann action

The local factors and actions of all blocks are extracted mechanically in exact rational arithmetic by the pipeline of Ref. [1], Eqs. (51)–(58), validated term by term on the closed-form interaction action of Ref. [2][1]. For the coin-cube conversion blocks the physical (Givens-lift) gauge yields an axis-universal action: 40 terms with identical degree structure for all three axes; the quadratic part is the signed-swap bilinear of C_a plus the environment transport term, and the environment control appears as quartic and higher couplings between the channel bilinears and $e'\tilde{e}$ [MACHINE-CHECKED]. Axis universality is a property of the physical gauge specifically: the bare (all-plus sign) gauge yields axis-dependent term counts (49/42/51)—discarding the lift signs breaks the relabeling equivalence, a gauge artifact asserted as such [MACHINE-CHECKED]. The streaming layers contribute the standard quadratic transport bilinears; one fresh-tape batch's lift equals the composition of its three single layers sign for sign, and its action is again a single transport bilinear of the composed permutation [MACHINE-CHECKED].

VIII. MASS: INVERSION DOUBLING

Doubling the coin with a mass bit b_m whose value *reverses all direction assignments*, $d_a^{(8)} = d_a \otimes \mathrm{diag}(1,-1)$, with conversions blind to b_m, places two opposite-chirality copies of the node at the same momentum and frequency (the b_m=1 sector is the spatial-inversion image; sector chiralities ∓ 1 verified) [MACHINE-CHECKED]. The alternative doubling $C_a \otimes Z$ fails: $-C_a$ generates the opposite triple orientation, which is the *other* corner frequency family, so the two sectors share no node. A mass layer $M = (1-q_m)\mathbb{1} + q_m C_m$ with $C_m = \mathbb{1}_4 \otimes XZ$ (a b_m-flipping controlled Givens, driven by a fourth environment field of density q_m, streamed at three phase-continuing swaps per cycle along a fixed axis so that its reads are also fresh—a once-per-cycle field outruns the carrier iff its speed is at least 3, and the fresh-tape identity then covers the massive model; on the torus the mass tape advances three sites per cycle against a carrier bound of two, so its first wrap re-read requires a cycle separation of $\lceil L/5 \rceil = 10$ at $L = 48$, beyond the $T = 6$ window used; the odd per-cycle batch count makes the massive and interacting rules homogeneous with period two in time, the free rule remaining period one) couples the chiralities. Two operator identities carry the sector [PROVEN]: $U(k) = (\mathbb{1} \otimes R_m)[U_4(k) \oplus U_4(-k)]$ and, at the node, $U(0) = U_4(0) \otimes M_2$ with M_2 a scaled rotation of the mass bit; hence the gap is *exactly*

$$2m \;=\; 2\arctan\frac{q_m}{1-q_m}, \qquad (7)$$

independent of q, with the multiplet center unshifted; frequencies add angles as in a telegraph process, and $m = q_m + O(q_m^2)$ recovers the Kac form [7]. On the node space the generators obey the full Dirac–Clifford algebra $\{h_a, h_b\} = 2v^2\delta_{ab}$, $\{h_a, \beta\} = 0$, $\beta^2 = 1$, identically in q [PROVEN], so $\omega = \omega_c \pm \sqrt{m^2 + v^2k^2}$ at leading order with the finite-k composition law $\cos(\omega - \omega_c) = \cos m \cos\varphi_{\mathrm{massless}}(k)$, φ the massless branch phase—accurate to 8×10^{-4} at $|k| = 0.15$, the residual being the inter-sector spinor mismatch [MACHINE-CHECKED]. No real Dirac mass exists at four components: the anticommutant of the quaternionic triple $\{C_a\}$ in $M_4(\mathbb{R})$ is zero-dimensional [MACHINE-CHECKED] (for a $\mathrm{Cl}(3,0)$ triple with squares $+1$ the anticommutant is two-dimensional with strictly negative squares— wrong-sign masses only); eight components are minimal, as realized here.

The census of Sec. VI C has a sharp massive fate, charted by the same sweep at $q \in \{0.08, 0.15\}$, $q_m = 0.05$ [MACHINE-CHECKED]. The mass layer breaks all axis permutations (exactly the 8 diagonal sign flips survive as spectral symmetries) and makes the spectrum self-conjugate at every k, so both quasienergy branches live at the same momenta. The *resonant principal quartet*—the corner and the three half-points, the self-conjugate nodes with $2k \equiv 0 \pmod{\pi}$, where $\omega \equiv -\omega$—is gapped by exactly $2\arctan[q_m/(1-q_m)]$ (two exactly two-fold levels split by $2m$ to 10^{-15}). *Every one of the 40 extra Weyl points survives exactly ungapped*: for each, the original doublet at ω and its inversion partner at $-\omega$ re-close at a common point (coinciding to 6×10^{-14}, displaced below 0.005π from the massless location) with unchanged integer charges. The mechanism is off-resonance: inversion doubling places the partner at the conjugate quasienergy $-\omega \neq \omega$, and the mass layer gaps only resonant sector pairs. Survivor charges sum to zero per branch. The massive model also adds structures with no massless counterpart: degeneracy curves confined to the $\{k_a \in \{0, \pi/2\}\}$ mirror planes (where the inter-sector mass matrix element vanishes by symmetry), isolated Weyl nodes pinned at the resonant quasienergies $\omega \in \{0, \pi\}$ (16 events at $q = 0.08$, charges summing to zero), and a 12-line edge network that remains exactly degenerate, with the original line doublet split only weakly ($2m/9.8$ at $(0.7, \pi/2, \pi/2)$, $q = 0.08$), leaving sub-

[1] The validation surfaced one typographical error in Ref. [2]: the coefficient of the top monomial in the closed-form interaction action, Eq. (38) there (source label CS9; equation number confirmed by compiling the arXiv source), must be -2, not -1; the printed value fails the defining relation $e^{-L} = K$ at that single monomial. The recomputation is committed as `wetterich_eq38_check` [MACHINE-CHECKED].

gap remnants. The massive model is therefore a Dirac quartet embedded in a spectator population that stays massless.

Inversion doubling also cancels the chiral anisotropy of Sec. VI B: the term is odd under the inversion that relates the two sectors, and the Dirac branches are symmetric functions of the sector pair, so on the doubled model every single-axis sign flip leaves the branch dispersion invariant to machine precision—below 10^{-15} at every radius to $|k| = 1.2$ and every mass density tested, against a relative splitting difference rising through 13% at $|k| = 0.02$ and 41% at $|k| = 0.08$ for the undoubled model; the transposition-chiral remnant is 8×10^{-4} at $|\delta k| = 0.08$ [MACHINE-CHECKED]. What survives is even in k: the family-ordered curvature spread of Fig. 3 is the even remnant of the same term, exact for the automaton; its magnitude sets the remaining distance to a Lorentz-symmetric dispersion (open problem (iii), Sec. XI).

Measured on the quenched vacuum with the gated eight-launch instrument (Fig. 3): the gate row returns $m = 0.0522 \pm 0.0056$ on the exactly known operator (closed form 0.0526; the replication's gate row returns 0.0569 ± 0.0026 on the same operator, so the half-range estimator's upward bias is of the size of the statistical error and the mass central values are bias limited), and the quenched row returns $m = 0.0569 \pm 0.0040$ ($+1.1\sigma$; the gap estimator is a half-range over noise-split pole phases and is upward biased under noise, to which part of this excursion is attributable; an independent-seed replication at doubled R returns 0.0557 ± 0.0041, $+0.8\sigma$) with multiplet center $\omega_c = 0.3224 \pm 0.0103$—consistent with the closed forms within statistics, as Theorem 3 extended to the massive model demands—and the dispersion lies on the exact operator branches with the family ordering of the curvatures that the operator prescribes [MEASURED].

IX. INTERACTION

For fixed media the multi-carrier sector is exactly determinantal (Sec. IV), so genuine interaction requires *back-reaction.* The imprint layer L4 acts once per cycle: at sites where an autonomous four-valued field fires (pair (xy), (yz) or (zx) with probability $g/3$ each) *and* the site's carrier fermion parity is odd, the selected two environment species at that site are swapped. The imprint field is autonomous in the same sense as the media: its randomness lives in the initial product measure only (value 0 with probability $1-g$; the four values are carried as two bits), and it evolves by legal deterministic pair-swap streaming at three phase-continuing swaps per cycle along a fixed axis—a once-per-cycle field outruns the carrier iff its speed is at least 3 (a rotating-axis schedule would need at least 7)—lifting with the standard reordering parity like every environment layer. A per-cycle resampled field would be a stochastic rule—an annealed approximation, not an automaton—and is not used. Three design facts govern the layer. (i) A fixed swap *pair* breaks cubic symmetry at $O(g)$ (anisotropic node corrections at 5.9σ and 12σ in the two independent channels) [MEASURED]; the rotating triple is protected. In the model defined here the protection is complete and needs no symmetry argument: the interaction law, Eq. (8) below, makes the interacting cycle operator a *scalar* multiple of the free one at every cycle, so every spectral property—the isotropy of the leading cone and the anisotropy structure of Sec. VI B alike—is independent of g [PROVEN]. (Under generic schedules, where the law holds only at leading order, full quenched A_4 covariance is *not* established: the double axis reversals are verified quenched per mode *including* the imprint back-reaction [MACHINE-CHECKED], cyclic axis rotations are verified at the annealed-operator level [MACHINE-CHECKED], single-axis reversals are provably not symmetries—no signed relabeling exists among all 384—and the measured interacting node isotropy under the generic schedule is constrained only at the few-percent level. None of this is needed for the model of this paper.) (ii) Parity control (not occupancy) is particle–hole even, so the complex structure of Sec. VII survives at $g > 0$ unchanged [MACHINE-CHECKED]. (iii) The lift gauge of the imprint is *physical*: the permutation lift ($|11\rangle \to -|11\rangle$) and the Givens lift differ by a parity-controlled sign with identical classical dynamics but per-event coherent deficits $2q^2$ versus $2q(1-q)$; we fix the permutation lift (minimal decoherence) [MACHINE-CHECKED].

Certificates. The connected two-particle correlator $C_2 = G_2 - \det G_1$ is nonzero already at $g = 0$—the model is an interacting carrier–environment theory at the correlator level, with free-fermion structure only per medium realization—so the interaction certificate is differential: $\Delta C_2(g) = C_2(g) - C_2(0)$ vanishes identically at $g = 0$ and switches on linearly, $\Delta C_2/g \simeq 0.35$ on the exact substrate (a one-dimensional ring with a fixed swap pair and a static imprint field—an interaction certificate, not the production three-dimensional layer) [MACHINE-CHECKED]. A control-variant experiment (occupancy-controlled imprint, used only as an instrument control) separates the parity-blocking contact artifact: 2–3% of the signal [MACHINE-CHECKED].

Write events and the flush. At $g > 0$ the imprint swaps two environment bits at the carrier's site—a *write*—and its permutation-lift sign reads both bit values into the amplitude. Freshness must therefore cover writes. The read→write direction is closed by the streaming alone (between any conversion read and a later write the bit has outrun the carrier: writes only ever touch never-read bits, so every lift sign-read is fresh), but a written or sign-read bit could be consulted at the very next read. The model therefore includes one post-imprint *flush* batch per environment field each cycle (the same certified layer type, origins continuing the field's phase), after which every written bit moves at least three sites before any field is read again: all reads are fresh at all times, and each cycle's imprint contributes the exact scalar $(1-g) + g(1-2q^2)$, independently of every-

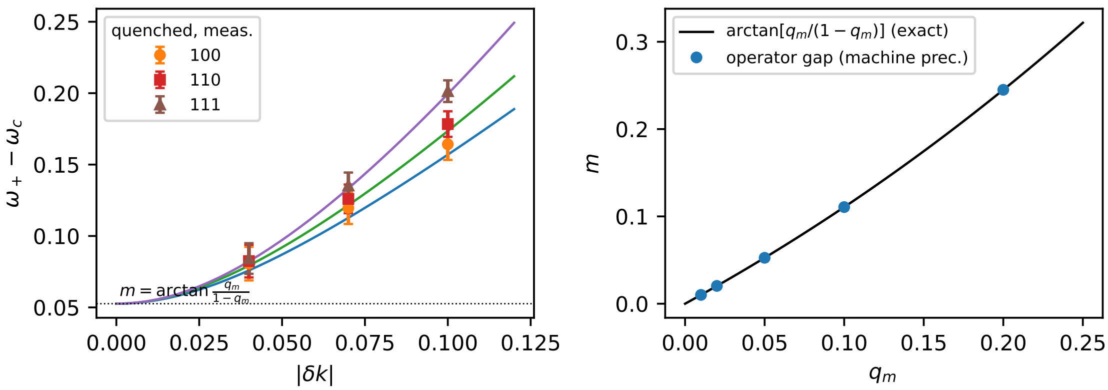

FIG. 3. Left: measured quenched massive dispersion (markers, $q = 0.08$, $q_m = 0.05$) on the exact operator branches (lines); the family ordering of the curvatures is the exact model's. Right: the operator gap against the closed form, Eq. (7), agreeing to machine precision. [MEASURED], [PROVEN]

thing else. The parity control makes the vacuum imprint-transparent, and

$$U_g(k) = (1 - 2gq^2)\, U(k) \qquad \text{exactly, at every cycle} \quad (8)$$

[PROVEN]: the cone, residues and mass are untouched, at a damping cost $\Gamma_{\text{int}} = 2q^2 g$ per cycle ($\sim 0.3\%$ of the free width at $q = 0.08$, $g = 0.1$). Machine check, exact at $T = 3$: all 2^{18} conversion histories $\times 4^3$ imprint outcomes (1.7×10^7 leaves), bits tracked through the streaming *and* through the write swaps, lift signs integrated exactly— $G_{\text{quenched}}(t) = (1 - 2gq^2)^t\, G_{\text{annealed}}(t)$ endpoint by endpoint to 2.1×10^{-15}, with zero correlated-read events of any kind; without the flush the law is exact at the first cycle but fails from the second through the identified sign-read channel (maximum endpoint deviation 1.1×10^{-3}), which is why the flush is part of the model [MACHINE-CHECKED]. One finite-size scope applies: with the flush each environment field advances nine sites per cycle—three batches, two from its own axis substeps and one from the flush—against a carrier bounded by two sites per cycle along the streaming axis, so the free-model horizon $T \leq \lceil L/8 \rceil$ does not transfer. The closing speed 11 gives $L > 11T + 6$ as a sufficient condition, and enumerating the wrap condition on the real substep word sharpens it: for $T \leq 12$ no wrap re-read exists once $L > 11T - 8$, the largest wrap-admitting size being 22 at $T = 3$ [MACHINE-CHECKED]. Both computations that invoke the law on a torus clear it with margin (the exact enumeration at $L = 48$ and the Monte Carlo run at $L = 40$, both $T = 3$). The identity that anchors the measurement: the media-ensemble walker average *equals* the canonical coherent-vacuum correlator exactly [MACHINE-CHECKED]. At Monte Carlo scale, paired common-noise walkers (exact $g = 0$ gate: the branches agree bit for bit) confirm the law at every retained cycle within the flip-shot-noise tolerance, with momentum spread at the noise floor ($\leq 4 \times 10^{-4}$ wherever all probe momenta are retained; at the last cycle the amplitude gate retains a single probe momentum). At these parameters the shot-noise tolerance exceeds the law's total deviation from unity, so the run's measured content is the isotropy and the per-cycle consistency; the law's *magnitude* rests on the exact enumeration above—isotropy survives interaction exactly, by theorem and by measurement [MEASURED].

X. DAMPING: A THEOREM AND ITS UNITARY COMPLETION

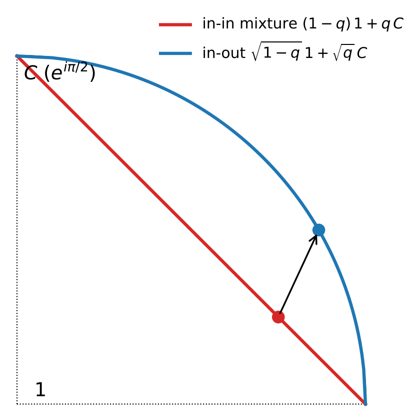

FIG. 4. Geometry of Theorem 4. Probabilistic (in-state) mixtures of the identity and a quaternion unit lie on the chord (damped interior); the two-boundary weights lie on the unitary arc. The isotropic node splitting is a property of the whole family $E_a(\alpha\mathbb{1} + \beta C_a)$, for all weights.

The dressed quasiparticle of the classical in-state observable is damped: $\Gamma(q) = -3\ln[(1-q)^2 + q^2]$ exactly, with quality factor $Q = \omega_0/\Gamma$ between 0.60 and 0.90 across the working range, *not* improving as $q \to 0$ inside the relativistic window (both ω_0 and Γ scale with q), and bounded at every density by Proposition 2 below. Proposition 1 states precisely what kind of damping this is: $U(k) = \rho^6 V(k)$ with $V(k)$ unitary, so $\Gamma = -3\ln\rho^2$ is a mode-blind, k-independent *interference-visibility* decay—every coherent mode decays at the same rate relative to the conserved probability sector, whose Perron eigenvalue in the doubled description remains exactly 1. It is not a broadening of the node relative to any other mode. Nor is it a removable normalization: the in-state normalization is fixed by probability conservation, and the visibility-to-probability ratio is physical. In the fresh-tape model there is no mode-selective decoherence at all (Theorem 3); under generic schedules it resides in the re-read corrections (Sec. V A). The overdamping is global, not a property of the working density:

Proposition 2 (Global overdamping) *At $k = 0$ each sub-step factor is ρ times a unit quaternion of angle $\theta(q) = \arctan[q/(1-q)]$, and the cycle is ρ^6 times left multiplication by their product; the bi-invariant triangle inequality for rotation angles on unit quaternions gives $\omega_0 \le 6\,\theta(q)$, hence*

$$Q \;=\; \frac{\omega_0}{\Gamma} \;\le\; \frac{2\,\theta(q)}{-\ln\rho^2(q)} \;\le\; \frac{\pi}{2\ln 2} \;=\; \kappa \qquad (9)$$

at every density, the last step because the middle bound is increasing in q with value κ at $q = \frac{1}{2}$. [PROVEN] *The model's actual supremum is sharper: $Q(q)$ increases monotonically with $\sup_q Q = \pi/(3\ln 2) = 1.511$ approached as $q \to \frac{1}{2}$* [MACHINE-CHECKED]*, so the in-state excitation retains at most $e^{-2\pi/Q} \le 2^{-6}$ of its visibility per oscillation period, at every density ($Q(0.08) = 0.66$).*

That the visibility decay cannot be tuned away is structural:

Theorem 4 (Q-pinning, quaternionic event class) *Let every conversion event lie in the quaternionic set $\mathbb{R}\cdot Q_8 = \{\pm\mathbb{1}, \pm C_x, \pm C_y, \pm C_z\}$. Then for any product of convex mixtures of events—non-commuting factors and arbitrarily correlated weights included—every eigenvalue λ of the composite obeys $\operatorname{dist}(\arg\lambda, (\pi/2)\mathbb{Z}) \le \kappa\,(-\ln|\lambda|)$ with $\kappa = \pi/(2\ln 2)$ exactly. Moreover, under three further hypotheses—(i) events in $\mathbb{R}\cdot Q_8$ controlled by per-engagement reads of binary media; (ii) the generic-schedule read geometry, in which the two same-axis engagements read lattice-adjacent sites with both offsets realized across coin branches; (iii) a deterministic environment rule with randomness confined to the initial measure—an isotropic node forces $\Gamma > 0$: node unitarity requires an almost-surely deterministic conversion word; among media factorizing over axis lines, the media with deterministic read-window counts are exactly the per-line crystal products, and every member destroys the cone.* [PROVEN]

(For the fresh-tape coincube itself no such hypotheses are needed: $\Gamma = -3\ln\rho^2 > 0$ holds exactly by Proposition 1 and Theorem 3; the second statement addresses whether *any* read geometry and medium in the class could do better, and answers no within its hypotheses.)

The restriction to the quaternionic class is necessary: for transposition events on three or more modes (eigenvalue angles $\{0,\pi\}$) products of mixtures develop three-cycle eigenvalues near $e^{2\pi i/3}$, and the phase-per-decay ratio is unbounded—for the product $[(1-\epsilon)S_{12} + \epsilon\mathbb{1}][(1-\epsilon)S_{23} + \epsilon\mathbb{1}]$ of two transposition mixtures it is 4.95 at $\epsilon = 0.05$ and 131 at $\epsilon = 0.002$; for event sets with finest eigenvalue angle $2\pi/\ell$ the constant grows as $\kappa(\ell) \to 2\ell/\pi$ [MACHINE-CHECKED]. The bound is a statement about the quaternionic class, degrading linearly with coin refinement. In these terms the theorem reads: no legal in-state automaton of this class can hold its node quasienergy farther than $\kappa\Gamma$ from $(\pi/2)\mathbb{Z}$. The bound governs the node—momenta where the transport phases vanish, so the composite stays inside the group algebra; at generic k the momentum factors $E_a(k)$ leave the event class and the ratio exceeds κ already on the coincube operator (reaching 6.06 at $q = 0.02$, $k = 0.4\,e_x$) [MACHINE-CHECKED]. And "an isotropic node forces $\Gamma > 0$" asserts a nonzero uniform visibility decay, not a node width.

The media hypothesis of the second statement is likewise scoped: for general three-dimensional translation-invariant media the deterministic-count classification is open—a family staggered along the read axis with an independent phase per transverse line lies outside the crystal list. It does not rescue the cone. Cross-streaming makes a carrier's two same-axis reads sample *distinct* transverse lines (adjacent under the generic schedule; the argument needs only distinctness), so per-line determinism does not give ensemble determinism: for independent phases of density r the ensemble one-cycle operator is damped by exactly $\Gamma = -3\ln[r^2 + (1-r)^2]$ per cycle, vanishing only at the crystal limits $r \in \{0,1\}$; and an exhaustive supercell construction (all 4096 members of the family on the 2^3 supercell, each exactly unitary and closed under the model's streaming) finds every one of its 11 864 degenerate band groups failing an isotropic-pair witness at the folded corner momenta—a witness that does detect the unitary-arc cone when fed one. Interior supercell momenta are not exhaustively excluded. [MACHINE-CHECKED]

The idealized ordered-dimer environment does achieve the commutator-limited decoherence $\Gamma = k^2/2$ per event—the branch-remerging mechanism is real—but its velocity fan collapses to the diamond: within the legal class, cone and in-state unitarity are mutually exclusive. Within the quaternionic class, *an isotropic cone forces overdamping*—$\Gamma > 0$ with $Q \le \pi/(2\ln 2)$ at every density (Proposition 2): the excitation always retains less than 2^{-6} of its visibility per oscillation period, an e-fold lost per tenth of an oscillation at the working density—*and in-state unitarity forces factorized transport*; both halves are proven above. The postselected two-boundary family below is the dichotomy's only known evasion; under the fresh tape it is exact at all times.

The evasion is the two-boundary object—a *postselected* amplitude, fixed by a uniform final environment boundary in the transfer-matrix formalism, not the evolution of a prepared state; whether the induced two-boundary weight is non-negative, as a probabilistic-automaton observable requires, is not addressed here and joins the open problems. The isotropic node splitting holds for the whole family $E_a(\alpha\mathbb{1} + \beta C_a)$, and by the identity of Proposition 1, $T^\dagger T = (\alpha^2 + \beta^2)\mathbb{1}$ for every real weight pair (the two-boundary walk is the $\alpha^2+\beta^2 = 1$ member of the same family): *every* product final boundary (w_0, w_1) yields zero relative damping and an isotropic node, landing on the unitary arc at $s^2 = w_1^2 q/[w_0^2(1-q) + w_1^2 q]$ [PROVEN]. The completions thus form a one-parameter family; in-state weights $(1-q, q)$ lie on the damped chord, and the square-root point $(\sqrt{1-q}, \sqrt{q})$ (Fig. 4) is selected uniquely as the only *medium-blind* member: among product final boundaries, $w_0 = w_1$ is the unique choice invariant under the medium's bit-flip involution—a criterion on the boundary weights alone—and the $\sqrt{q}$ structure of the induced transfer pair is its consequence, not its definition [PROVEN]. Numerically: all $|\lambda| = 1$ to 4×10^{-16} with slope isotropy asserted below 10^{-3} and ω_0 tunable at the $\sqrt{q}$ scale [MACHINE-CHECKED]. In the fresh-tape model the two-boundary amplitude inherits the no-re-read property of Theorem 3 (the same path sum, different boundary weights), so the in–out family is exactly unitary at *all* times—machine check: the exact quenched in–out path sum under the fresh schedule equals the arc-operator power to 1.6×10^{-15} at every cycle ($L = 24$, $T = 3$, three momenta) [MACHINE-CHECKED]; under generic schedules the fresh-read law is exact at the first cycle only, with re-read corrections growing to 11–24% by three cycles at $q = 0.08$. What remains open is not the survival of the object but its interpretation: the positivity of the induced two-boundary weight (Sec. XI). [PROVEN], [MACHINE-CHECKED]

XI. DISCUSSION

Under the strictest single-fermion contract, the construction half of the three-dimensional problem posed in Ref. [1] is solved exactly: a legal probabilistic cellular automaton whose dynamics is exactly unitary quantum mechanics, whose quenched single-particle propagator *is* its Bloch operator theory (Theorem 3), and whose single-carrier excitation on a clustering product vacuum is a Weyl fermion with an exactly isotropic leading cone—dispersion and spinor residues, exact by theorem and confirmed by gated measurement—extendable to a massive Dirac quartet with an exact tunable mass, and to a certified media-mediated interaction whose damping law is exact at every cycle. The Lorentz-symmetric half of the problem is delivered at leading order only: beyond it stands the structural chiral anisotropy of Sec. VI B, whose mechanism, exhaustive unavoidability, and partial remedies this paper charts. Every closed form is a statement about the automaton itself, every measurement runs behind a known-answer gate, and the estimator's anisotropy resolution is established by positive controls. The solution is a lattice fermion with a computed spectator census: the sweep of Sec. VI C places 43 further Weyl points and three nodal lines at momentum distance $\geq 0.41\pi$ from the working node, and the massive model gaps only the resonant quartet among them.

The precise open problems are: (i) the half-filled carrier sector of the complex-structure construction in three dimensions; (ii) the positivity of the induced two-boundary weight (its long-time survival is settled by Theorem 3; re-read resummation remains relevant only for generic schedules); (iii) the Lorentz-symmetric continuum limit: the chiral $O(|k|)$ anisotropy is structural at four components (Sec. VI B), cancels at first order only on the inversion-doubled branch, and is reduced but not removed by the legal counterterm family (45%) and by density optimization ($c \propto 1/q$)—an exact cancellation mechanism, or an enlarged construction class in which the term is not tied to the net chirality, is the open problem, interaction corrections beyond the scalar law included; and (iv) multi-carrier physics beyond the contact terms already present in the extracted quartic action; and (v) the spectator population of the massive census—the exactly massless off-resonant Weyl species, the pinned nodes, and the sub-gap nodal-line remnants—and whether a legal layer can gap it: the octet pair persists with constant opposite chiralities across the entire density range, separated in both momentum and quasienergy, so no static translation-invariant perturbation suffices except near $q \approx 0.339$ and 0.370, where opposite-chirality quasienergy crossings open momentum-only bridges [MACHINE-CHECKED]. The damping of the in-state excitation is not on this list: it is an exact, mode-blind prediction of the construction (Proposition 1) of stated magnitude—$Q \approx 0.66$ at the working density, $Q \leq \pi/(2\ln 2)$ at every density—and the dichotomy, cone forces overdamping (Proposition 2) while in-state unitarity forces the diamond (Theorem 4), stands as the second principal result of this paper, with the postselected two-boundary family as its only known evasion, exact at all times under the fresh tape.

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Appendix A: Proofs

1. Theorem 1 (finite rays)

Let the rule conserve particle number in the one-particle sector. Its restriction to that sector is a signed permutation of the species basis composed with site shifts, so the Bloch operator has matrix elements $S(k)_{\alpha\beta} = s_{\alpha\beta} e^{ik\cdot d_{\alpha\beta}}$ with, by the unique-jump property, exactly one nonzero entry in each row and column: a monomial matrix. Its characteristic polynomial factorizes over the cycles of the underlying permutation; a cycle of length ℓ, accumulated displacement D and sign σ contributes $\lambda^\ell = \sigma\, e^{ik\cdot D}$, i.e. $\omega_m(k) = (k\cdot D + \arg\sigma + 2\pi m)/\ell$: every branch is exactly linear with group velocity D/ℓ, drawn from the finite set of cycle data. A cone $\omega = v|k|$ requires a two-sphere of group velocities. The deviation of any finite ray set from a sphere is scale invariant, so no continuum limit removes it; and a periodic cycle of rotated rules is again unique-jump (products of signed permutations are signed permutations), so the theorem applies to the composite rule. □

2. Theorem 2 and the accompanying search

Three ingredients; the theorem is (ii), the rest is the search. (i) *Algebraic ceiling.* On $\mathfrak{sl}_2(\mathbb{R})$ every element satisfies $X^2 = -\det(X)\mathbb{1}$ and anticommuting elements are orthogonal in the polarization of det; in the basis $\{X, Z, XZ\}$ the squares are $+\mathbb{1}, +\mathbb{1}, -\mathbb{1}$, so the form has signature $(2, 1)$ and no real two-dimensional representation of $\mathrm{Cl}(3, 0)$ exists. (ii) *Generic weights.* A symbolic orthogonality lemma for the mixture families shows that a degenerate propagating pair at any momentum requires $(1-p)^2 = p^2$ for rotation coins, is impossible for reflection coins, and degenerates for diagonal coins: a linear node forces $U(k_0) = \lambda_0\mathbb{1}$, the splitting generators are conjugated involutions with automatically scalar anticommutators (2×2 Cayley–Hamilton), and the required bilinear orthogonality gives $T_{00}T_{11} = -T_{01}T_{10}$. For the blocked two-origin cycles the exclusion is exhaustive in the tables and heuristic in momentum: all 512 sign/direction designs per (placement, engagement, p) sector are enumerated, with $p \in \{0.15, 0.25, 0.35, 0.5\}$ and 14-start continuous momentum solving, and every design carries an explicit certificate—a candidate node entering the sector's anisotropy floor, a degenerate-only scalar point, or no scalar point at all, the last certified by a minimum scalar residual ≥ 0.074 against the 10^{-9} solver tolerance of solvable designs. The floors are claimed at the sampled grid values of p only; between them they dip toward zero on approach to the tuned p^*, which is the content of part (iii), not a violation. (iii) *The fine-tuned points.* At the singular weight $p = \frac{1}{2}$ the single-origin placement has $U(k_0)$ scalar at $k_0 = (\frac{3\pi}{4}, \frac{3\pi}{4}, -\frac{\pi}{4})$ (and its symmetry images), and the three effective generators form an exact complex Clifford triple, giving $\omega = \pi \pm |\delta k|$: an isotropic node with every parameter frozen and $\Gamma = \frac{3}{2}\ln 2$ at the chord midpoint. In the blocked placements, branch continuation in p (warm-started exact scalar solves with probe-radius-extrapolated anisotropy) locates codimension-one transversal zeros of the single remaining Gram condition at $p^* = 0.3300$ (additive placement, $k_0/\pi = \pm(0.916, 0.916, -0.916)$, $|\lambda_0| = 0.1736$, $\chi = -1$) and $p^* = 0.3204$ (multiplicative, $|\lambda_0| = 0.1799$, $\chi = +1$), both with $\omega_0 = \pi$: isolated tuned nodes, not parameter families. All parts are machine-checked, the sweep exhaustively. □

3. Theorem 3 (fresh tape)

Lemma A. A pair-swap layer with origin o moves the bit at 1D site y to $y + 1$ if $y \equiv o \pmod 2$, else to $y - 1$ (finite check, four cases).

Lemma B (monotone transport). Under any origin-alternating swap sequence with phase p (the k-th swap has origin $(p + k) \bmod 2$), the class $\epsilon = (-1)^{y+p}$ of a bit is conserved and its displacement after k swaps is exactly ϵk: by Lemma A the bit moves $+1$ iff its position parity matches the current origin, and both flip at each step, so the direction is permanent. (A schedule restarting each batch at origin 0 breaks the alternation at the batch boundary; its transport is 0 per batch—the machine-checked negative control.)

Lemma C (slot counting). Relative to the axis-a field the cycle is the periodic word $AASSXX$ (A: axis-a substep—the path reads the field at its own site, then the field streams one batch; S: axis-s_a substep; X: third axis). Between two A-slots separated by Δn batches the path takes at most $m \leq \Delta n + 1$ s_a-steps, and when Δn is even, $m = \Delta n$ (exhaustive over 12 cycles).

Proof of (a). Pair-swap layers permute the axis-a field only along s_a, so a re-read at substeps $\tau_1 < \tau_2$ requires zero net path displacement along both transverse axes and equal path and bit displacements along s_a. By Lemma B the bit moves exactly $3\Delta n$ sites in a fixed direction; by Lemma C the path moves at most $\Delta n + 1$. Since $3\Delta n > \Delta n + 1$ for every $\Delta n \geq 1$, the match fails at every separation—no residual window. As an independent check that the reduction models the automaton faithfully, the exact path enumerations (labels tracked through the real permutations, all signs and weights) cover every separation $\Delta n \leq 7$ and find zero re-reads with $G_{\text{quenched}} = G_{\text{annealed}}$ to 10^{-16}.

Proof of (b) and sharpness. On the ring of circumference L the match condition becomes $\epsilon\, 3\Delta n \equiv \Delta y \pmod L$ with $|\Delta y| \leq m$ and $\Delta y \equiv m \pmod 2$; transverse return along axis a forces Δn even, and then $m = \Delta n$ by Lemma C. The first solution is $\Delta n^* = 2\lceil L/8\rceil$, so an evolution of T cycles is re-read-free iff $2T - 2 < 2\lceil L/8\rceil$, i.e. iff $T \leq \lceil L/8\rceil$. The certificate reproduces Δn^* from the real tapes at $L = 12, 16, 24, 32$, and the full path enumeration at $L = 16$ exhibits the first wrap re-reads at $T = 3 = \lceil 16/8\rceil + 1$ with minimal separation exactly

4 batches. $\square$

4. Particle–hole equivariance, $[\hat{S}, P] = 0$

Lemma (string factorization). Let B be a parity-even operator with support T, and T_c the carrier modes in T. The Majorana factors of P anticommute pairwise, so sorting gives $P = \sigma\, P_{T_c} P_{T_c^{\complement}}$ with a fixed reordering sign $\sigma = \pm 1$; a parity-even B commutes with every Majorana off its support, hence with $P_{T_c^{\complement}}$, and $[B, P] = \sigma\, [B, P_{T_c}]\, P_{T_c^{\complement}}$: the block-local check suffices.

Layers. L2 acts, per fired site, on the four same-site carrier modes (plus the control); the stored Jordan–Wigner spectators lie inside the contiguous block, so $[\mathrm{L2}, P] = 0$ reduces to one finite block check per axis, independent of L. L1 is a canonical permutation lift, for which $U_\pi P U_\pi^{-1} = \mathrm{sign}(\pi)\, P$; per channel the axis-$a$ shift is L^2 disjoint L-cycles, so $\mathrm{sign}(\pi) = (-1)^{(L-1)L^2} = +1$ for every L (even L: L^2 even; odd L: $L-1$ even). L3 is supported on environment modes and parity even, so it commutes with every carrier Majorana. L4's carrier part is a parity projector and its environment part a string-free adjacent swap, block-checked in both lift gauges. Size independence is structural: the same finite blocks recur at every site of every lattice, so the identity holds on the full Fock space at every L. $\square$

5. The mass identities, Eq. (7)

First, the factorization $U(k) = (\mathbb{1} \otimes R_m)[U_4(k) \oplus U_4(-k)]$ with $R_m = (1-q_m)\mathbb{1} + q_m XZ$: (1) $d^{(8)} = d \otimes \mathrm{diag}(1, -1)$, so every transport phase is block diagonal in b_m with opposite momentum sign in the two blocks, $E_8(k) = E_4(k) \oplus E_4(-k)$; (2) $C_8 = C_4 \otimes \mathbb{1}$, so conversions are block diagonal and identical in the two blocks; hence the pre-mass cycle is $U_4(k) \oplus U_4(-k)$ and the once-per-cycle mass layer is $\mathbb{1}_4 \otimes R_m$ (machine-checked symbolically in the full vector momentum). Second, the gap: at $k = 0$ the momentum factors are the identity, so the direction reversal that distinguishes the two b_m sectors is invisible and the axis layers act as $U_4(0) \otimes \mathbb{1}_2$. The mass layer acts on the b_m factor alone: $M_2 = (1-q_m)\mathbb{1} + q_m XZ$, a real multiple of a rotation with arg-eigenvalues $\pm \arctan[q_m/(1-q_m)]$ and modulus $\sqrt{(1-q_m)^2 + q_m^2}$. Hence $U(0) = U_4(0) \otimes M_2$: the spectrum is the product set, frequencies add angles, the multiplet center is unshifted, and the gap is exactly $2\arctan[q_m/(1-q_m)]$, independent of q. $\square$

6. The imprint amplitude law

In the one-particle sector the carrier's site has odd parity, so the imprint fires iff $\iota \neq 0$, probability g. The permutation lift of the fired pair swap has sign table $|00\rangle \to +|00\rangle$, $|01\rangle \to +|10\rangle$, $|10\rangle \to +|01\rangle$, $|11\rangle \to -|11\rangle$ (the canonical transposition lift), and the swap preserves the Bernoulli pair measure, so the per-event coherent factor is the plain sign average $(1-q)^2 + 2q(1-q) - q^2 = 1 - 2q^2$, identical for the three pairs. One line of expectation: $U_g = (1-g)U + g(1-2q^2)U = (1-2gq^2)U$ per cycle; under the fresh-tape model every read is fresh, so the law holds exactly at every cycle. The Givens lift's table gives $1 - 2q(1-q)$ instead: the law is lift-gauge dependent, the isotropy of the damping is not. $\square$

7. Proposition 2 (global overdamping)

At $k = 0$ the momentum factors are the identity and each sub-step factor is $(1-q)\mathbb{1} + qC_a = \rho\,[\cos\theta + \sin\theta\, C_a]$ with $\theta = \arctan[q/(1-q)]$: ρ times the left-regular representation of a unit quaternion of rotation angle θ. The cycle at $k = 0$ is therefore $\rho^6 L_u$ with $u = u_z^2 u_y^2 u_x^2$ a product of six unit quaternions of angle θ each, and its eigenphases are $\pm\Theta(u)$, the angle of u. The bi-invariant metric on the unit quaternions ($S^3 \cong SU(2)$, geodesic distance = rotation angle) satisfies the triangle inequality $\Theta(uv) \leq \Theta(u) + \Theta(v)$, so $\omega_0 = \Theta \leq 6\theta$. Dividing by $\Gamma = -3\ln\rho^2$ gives $Q \leq 2\theta/(-\ln\rho^2) =: g(q)$; g is increasing on $(0, \frac{1}{2}]$ (machine-checked densely) with $g(\frac{1}{2}) = (\pi/2)/\ln 2 = \kappa$, giving Eq. (9). The model's supremum $\pi/(3\ln 2)$ follows from $\omega_0 \to \pi$ and $\Gamma \to 3\ln 2$ as $q \to \frac{1}{2}$, with $Q(q)$ monotone on a dense grid [MACHINE-CHECKED]; the per-period statement is $e^{-2\pi/Q} \leq e^{-2\pi\cdot 3\ln 2/\pi} = 2^{-6}$. $\square$

8. Theorem 4 (Q-pinning)

Any product of convex mixtures of the event set is itself a convex mixture over the quaternion group Q_8 (the product measure is the convolution on the group, so correlations between events are automatically included). In the quaternion representation such a mixture is $T = w\mathbb{1} + xC_x + yC_y + zC_z$ with $|w| + \|(x,y,z)\|_1 \leq 1$, and its eigenvalues are $w \pm i\|(x,y,z)\|_2$. Since $\|\cdot\|_2 \leq \|\cdot\|_1$, every eigenvalue lies inside the convex hull of the chords between adjacent fourth roots of unity. On the extremal chord $1 \to i$, $\lambda(t) = (1-t) + it$, the maximized quantity is $\mathrm{dist}(\arg\lambda, (\pi/2)\mathbb{Z})/(-\ln|\lambda|)$: by the square-hull reduction and the $\pi/2$ rotation symmetry of the event angles it attains its maximum $\pi/(2\ln 2)$ at the chord midpoint $t = \frac{1}{2}$, where the distance folds from $\arg\lambda$ to $\pi/2 - \arg\lambda$. (The unfolded ratio $\arg\lambda/(-\ln|\lambda|)$ diverges toward the chord endpoint and is maximized nowhere interior; a dense hull scan confirms no interior point exceeds κ.) This bounds $\mathrm{dist}(\arg\lambda, (\pi/2)\mathbb{Z}) \leq \kappa\Gamma$ for arbitrary, non-commuting products—the full statement of the theorem. The group closure is the load-bearing hypothesis: outside $\mathbb{R}\cdot Q_8$ (already for transposition events on three modes) products of mixtures leave the span of the event set, three-cycle eigenvalues appear, and the measured phase-per-decay ratios grow without bound, as reported in the main text. For the second statement (under hypotheses (i)–(iii), the read-window structure being hypothesis (ii)): unitarity of the node eigenvalue requires the conversion count $N \pmod 4$ along a read window to be deterministic, by $|\mathbb{E}[i^N]|^2 = (p_0 - p_2)^2 + (p_1 - p_3)^2 \leq 1$ with equality only at a vertex; on rings, and on three-dimensional media factorizing over axis lines, the media with deterministic window counts are exactly the (per-line) crystals (exhaustively enumerated at $L = 6, 7, 8, 10$), with the non-factorizing phase-per-line family closed separately as described in the main text; and the deterministic-word cycles' spectra are computed explicitly and are never isotropic. Machine checks accompany every step, including randomized non-commuting products. □

Appendix B: Verification protocols

Every claim tagged [MACHINE-CHECKED] is an assertion in a runnable script; the tolerances quoted below are the assertion thresholds, and each script is self-checking (it fails loudly rather than reporting).

Layer lifts. The conversion layer is verified densely on the one-site Fock space (32 states): the controlled double Givens is a signed permutation of the basis, unitary, commutes with fermion parity, is the identity on the empty control sector, has single-particle matrix equal to the signed-swap block of C_a, satisfies $G^2 = -\mathbb{1}$ on the one-particle sector, and maps the doubly occupied pair to + itself (determinant one), for all three axes, to 10^{-12}.

Composite sign coherence. The full cycle is built as a signed permutation of the many-body basis from the stored per-layer sign rules (spectator Jordan–Wigner strings for conversions; inversion and wrap parity for translations), in the segregated mode ordering. For random environment configurations the zero-, one- and two-particle sectors are extracted and compared: the one-particle sector must equal the stated rule and the two-particle sector the antisymmetrized square $\Lambda^2 M_1$, exactly. This holds on the full Fock space of a ring substrate (2^{15} states) and in three-dimensional geometry at $L = 2$ (496 pair states) and $L = 3$ (5778 pair states), under both the fresh-tape and the generic streaming schedules. Mutation controls certify sensitivity: deleting the spectator string produces 8/496 mismatches; deleting translation parity produces 2772/5778 at $L = 3$ (counts are specific to the committed environment draw) and—the recorded caveat—0 at $L = 2$, where two sub-steps per axis make net crossings even.

Complex structure. P is applied as the Majorana string with its Jordan–Wigner signs; $[\hat{S}, P] = 0$ is checked as an identity of signed permutations (both orderings byte-identical), on the full substrate Fock space and, in three dimensions, on the sectors $\{0, 1, 2, 3, M-2, M-1, M\}$ (three-carrier wedges, 1.6×10^6 state–draw evaluations) using particle–hole duality to reduce the deep sectors to one- and two-hole problems; per layer and for the full cycle, under both streaming schedules. The unitarity of the complex picture is verified as invariance of the induced Hermitian form on random vectors to 10^{-9}.

Grassmann extraction. Local factors are built from the block tables over exact rationals; $e^{-L} = K$ is asserted exactly before any action is reported, and the inverse map $K \to (\mathrm{perm}, \mathrm{signs})$ must round-trip. The pipeline is validated term by term against the closed-form interaction action of Ref. [2].

Table II maps each principal claim to its verification script in Ref. [29].

TABLE II. Principal claims and their verification scripts (all self-asserting; in `scripts/` of Ref. [29] unless noted).

finite-ray tests	`tests/test_finite_ray_*`
$v(q)$ dressing law	`theory_linear_law`
Abelian-gauge diamond	`w2_scalar_gauges`
coin scans ($n = 2, 4$)	`w3_transfer_scan`
annealed cone	`w3_cone_verify`, `theory_real_cone`
lift certificates	`w3c_lift_check`
composite sign coherence (1D ring)	`w3c_composite_check`
3D sign coherence; sectors	`theory_3d_certs`
$[\hat{S}, P]$ proof blocks	`theory_sp_proof`
fresh-tape theorem	`freshtape_proof`
re-read kinematics	`reread_kinematics`
chiral anisotropy; no-go	`theory_anisotropy`
global overdamping	`theory_q_global`
fresh-tape schedule tests	`tests/test_freshtape`
quenched cone (gated)	`w3c_fresh`
generic-schedule cone/helicity	`w3c_corner`, `w4_helicity`
generic mass/interaction	`m8_corner`, `i3_quenched3d`
census, charges	`census_sweep`
factorization; structural loci	`spectrum_census`
positive controls	`w3c_positive_control`
positive control (fresh)	`w3c_control_fresh`
two-component taxonomy	`theory_redteam`
helicity residues	`w4_helicity_exact`, `w4_fresh`
complex structure	`e1_complex_structure`
substrate bridge	`bridge_check`
Grassmann actions	`e2_coincube_actions`
Eq. (38) erratum	`wetterich_eq38_check`
mass sector	`m8_mass_exact`, `m8_fresh`
mass structure	`theory_mass`
mass-gap identity	`theory_mass_identity`
interaction exact law	`freshtape_interaction`
interaction measured	`i3_fresh`
interaction certificates	`i2_connected`, `i2_split`
imprint-field theory	`theory_interaction`
spectator flow (open problem v)	`orbit_flow`
rationality of closed forms	`theory_rationality`
Q-pinning	`theory_q_bound`
in–out amplitudes	`a_inout`, `a_inout_3d`
in–out under fresh tape	`inout_fresh`

Appendix C: Measurement pipeline

All [MEASURED] results use one instrument family. The object is the exact per-medium single-particle signed field (the single-carrier sector is linear in the field for fixed media, so this is walker-exact with infinite walkers), averaged over R media; the media-ensemble average provably equals the canonical coherent-vacuum correlator.

Bloch-matrix estimation. All n basis launches are evolved, giving the full $n\times n$ propagator series $G(t,k)$; the one-cycle operator is fit by least squares, $\hat{U}(k) = G_{t+1}G_t^{+}$ over a post-transient window. Momenta are probed off the lattice grid by direct phase contraction, with probe radii inside the node's measured linear range.

Pooling. Naive eigenvalue extraction near a two-fold node is $\sqrt{\text{noise}}$-ill-conditioned (a noisy nearly defective matrix splits its double eigenvalue as the square root of the perturbation); pooling is therefore performed on characteristic-polynomial coefficients over the symmetry orbit of measurement momenta—a basis-invariant, *linear* (hence unbiased) average—before rooting. Slopes use symmetric splittings at two step sizes with $\delta k \to 0$ Richardson extrapolation. Media are accumulated in independent blocks (ten for the cone instrument, eight for the massive instrument, six for the helicity instrument) and every final quantity (slope, ratio) is jackknifed at the end-of-pipeline level over leave-one-block-out samples; the annealed gate row's deviation from the exact operator is recorded per channel and reported alongside as the instrument systematic—the quoted errors are the block jackknives. Off-symmetry direction families are measured alongside the cubic stars. The diamond-exclusion significance is computed in code for every channel by the generic-schedule instrument; for the fresh-model rows the quoted significances are the committed ratios' distances from the benchmark values in units of their own jackknife errors—direct arithmetic on the stored rows—with the weakest channel quoted. The three campaign instruments draw their media from a common base seed stream, so their quenched excursions are correlated across instruments; independent-seed replications at doubled R accompany the campaign: the cone replication reproduces the quenched node at -0.4% of the closed form and the mass replication returns the gap at $+0.8\sigma$.

Gates. Every production run carries an annealed known-answer row through the identical pipeline; a failed gate discards the run. The cone instrument gates both rows against the closed forms; the massive and helicity instruments gate the known-answer row only, so their quenched rows are protected by the gate row's validation of the pipeline rather than by a bound of their own. Amplitude gates exclude momenta near free-beat amplitude nodes for ratio estimators (shot-noise gating alone is insufficient there). Eigenvector work uses the $e^{+ik\cdot x}$ contraction (Sec. VI). For interaction ratios, paired common-noise evolution (identical randomness with the imprint on and off) cancels the medium-ensemble variance, which is otherwise dominated by low-order conversion caustics. Gate tolerances are set to catch catastrophic estimator failure, not to certify accuracy: the cone-ratio gates use $\max(2\%, 3\sigma)$ of the row's own jackknife, the fresh instruments gate both rows' node parameters at $\max(0.5\%, 3\sigma)$, the massive-gap gate allows 30% of the gap with the multiplet center gated at 0.02, and the generic instrument's node-modulus gate is 0.05 absolute. Accuracy statements never rest on gates; they rest on the quoted statistical errors and the recorded systematics, and the estimator's ability to resolve anisotropy is established by positive controls (a synthetic diamond-splitting truth and a legal broken-coin walk) run through the identical pipeline. A committed broken run under the generic schedule (probe radii outside the node's linear window) documents why the controls carry the burden: despite a 2.4× absolute-speed failure it returns cubic ratios of 1 at the 10^{-4} level—a ratio near 1 is, by itself, a weak discriminator.

Recorded failure modes. The following estimator pathologies were each encountered, diagnosed and firewalled during this program, and are listed because they generically afflict lattice pole measurements: (i) $\sqrt{\text{noise}}$ splitting at near-degenerate poles; (ii) exceptional-point misidentification in damped operators (real double eigenvalues accepted as propagating nodes); (iii) through-origin fits across a packet's death into the noise floor; (iv) ratio blowup at beat-amplitude nodes; (v) probing outside the (renormalized) linear cone window; (vi) branch mixing under scalar single-pole estimators at two-fold degenerate nodes; (vii) one-dimensional co-streaming substrates, whose companion locking invalidates absolute calibrations that do transfer in three dimensions.

Appendix D: Closed forms

TABLE III. Exact expressions for the model's free and dressed spectra (exact for the automaton by Theorem 3). All are verified against the operators they describe to at least 10^{-9}; $D = \sqrt{1-q} + \sqrt{q}$.

node modulus	$\lvert\lambda_0\rvert^2 = [(1-q)^2 + q^2]^6$
factorization	$U = \rho^6 V$, $\rho^2 = (1-q)^2 + q^2$, $V(k)$ unitary
free width	$\Gamma(q) = -3\ln[(1-q)^2 + q^2]$
cone speed	$v(q)$: $2/\sqrt{3} \to 1.207$ (max at $q \simeq 0.30$)
dressing law (1D)	$v = 2\lvert 1-2q\rvert$
mass gap	$2m = 2\arctan[q_m/(1-q_m)]$
mass-layer modulus	$\sqrt{(1-q_m)^2 + q_m^2}$ per cycle
massive dispersion	$\cos(\omega - \omega_c) = \cos m \cos\varphi(k)$
odd anisotropy	$s - 2v\lvert k\rvert = c(q)\, k_x k_y k_z / \lvert k\rvert$, $c(0.08) = 42.7$
interaction law	$U_g(k) = (1 - 2gq^2)\, U(k)$
Q-pinning constant	$\kappa = \pi/(2\ln 2) = 2.2662$
global Q bound	$Q \le 2\theta/(-\ln\rho^2) \le \kappa$; $\sup_q Q = \pi/(3\ln 2)$
in–out transfer	$E_a[\sqrt{1-q}\,\mathbb{1} + \sqrt{q}\, C_a]/D$, $\lvert\lambda\rvert \equiv 1/D$

Table III collects the closed forms. Two remarks. The in–out transfer's uniform modulus $1/D$ per read is a global scalar (every path reads one bit per sub-step) and is absorbed into wave-function renormalization; the physical statement is the vanishing of all *relative* damping. The quality factor of the in-state quasiparticle, $Q = \omega_0/\Gamma$, rises from 0.60 to 0.90 over $q \in [0.02, 0.25]$, monotonically to $\pi/(3\ln 2)$ at $q \to \frac{1}{2}$ (Proposition 2); the quantity Theorem 4 bounds by κ is $\mathrm{dist}(\omega_0, (\pi/2)\mathbb{Z})/\Gamma$, which coincides with Q for $q \lesssim 0.2$ (where $\omega_0 < \pi/4$).

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